

Pancham Vishwanath Kotian

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PROFESSIONAL SUMMARY

Hardware and embedded engineer with diverse experience. Spent one year as a co-founder developing an IoT product. Completed one year of freelance hardware design. Also brings four and a half years of enterprise software experience, with strong cross-functional collaboration skills. Skilled in PCB design using KiCad and EasyEDA. Proficient with microcontrollers such as STM32, ESP8266, NXP, and ATmega. Experienced in system debugging and simulation tools including LTSpice, ROS, and Python for control systems and sensor fusion. Ready to contribute immediately to R&D in Pune.

WORK EXPERIENCE

- **IndependentResearcher.in & other clients** Remote
Freelance Software & Embedded Engineer Feb 2026 – Present
 - Engineered physics-based simulations for web deployment; implemented Kalman filters, Gaussian noise reduction, and control loops—skills directly applied to embedded sensor processing.
 - Implemented harmonic analysis and stable control systems using ROS and PHP. Simulated a rover system in Unity and used Python for automation and scripting.
 - Crafted custom embedded solutions for various clients, handling everything from requirements to deployment.
- **Capgemini** Navi Mumbai, India
Senior Associate Consultant (B1) June 2022 – 2025
 - Architected and deployed automation solutions that reduced manual effort by 70% in enterprise workflows.
 - Mentored junior team members and coordinated with global stakeholders to deliver high-impact customizations.
 - Led L3 technical support for live ERP systems; resolved 50+ critical tickets per month through systematic debugging and root cause analysis.
- **Capgemini** Bangalore, India
Senior Analyst (A4) June 2020 – May 2022
 - Developed and implemented automation scripts that streamlined ERP processes, achieving 70% reduction in manual effort.
 - Collaborated with cross-functional teams to gather requirements and deliver customizations under tight deadlines.
 - Provided L2/L3 technical support; handled 40+ tickets monthly with high customer satisfaction.
- **Bitleaf** Dombivli, India
Co-founder & Hardware Designer 2019 – 2020
 - Spearheaded an end-to-end IoT product using ESP8266; handled PCB design (KiCad), power management (including PD charging with TPS25751), and firmware development.
 - Performed hardware bring-up, tested with oscilloscope and multimeter, and integrated with cloud platforms.
 - Managed vendor communication and prototyping, ensuring design manufacturability.

KEY PROJECTS

- **Sensor Interface PCB** 2020
Created a custom 2-layer PCB to interface environmental sensors with an STM32 microcontroller. Optimized layout for signal integrity and compact form factor; successfully ordered and tested prototypes. (Freelance project for BookPasskar.com)
Tools: EasyEDA, KiCad, I2C/SPI
- **IoT Product with PD Charging** 2019-2020
Built a connected device with power delivery support; integrated battery charging and wireless control. Implemented firmware for sensor data acquisition and MQTT communication.
Tools: ESP8266, KiCad, TPS25751, Embedded C
- **Control Systems & Sensor Fusion Simulations** 2026
Developed interactive simulations for harmonic analysis, stable control loops, and Kalman filter-based noise reduction. Used ROS to prototype sensor integration; demonstrated real-world applicability to embedded systems.
Tools: Python, ROS, LTSpice, Unity

EDUCATION

- **Datta Meghe College of Engineering** Airoli, Maharashtra, India
Bachelor of Engineering in Electronics 2019
 - First Class
- **Holy Angels Jr. College** Dombivli, Maharashtra, India
Higher Secondary Certificate (HSC) 2015
 - First Class
- **South Indian Association High School** Dombivli, Maharashtra, India
Secondary School Certificate (SSC) 2013
 - First Class

TECHNICAL SKILLS

- **Languages:** Python, Embedded C, SQL/PLSQL, Baan (4GL), HTML
- **Hardware Tools:** KiCad, EasyEDA, LTSpice, ROS, Unity, Git, Docker
- **Microcontrollers:** STM32, ESP8266, NXP (LPC), ATmega, PIC
- **Debugging & Test:** Oscilloscope, Multimeter, Logic Analyzer, ST-Link, J-Link compatible debuggers
- **Design & Simulation:** Schematic capture, PCB layout, control systems, Kalman filters, harmonic analysis
- **Operating Systems:** Ubuntu Linux, Microsoft Windows

LANGUAGES

- English (Fluent), Hindi (Fluent), Marathi (Fluent), Kannada (Conversational)

DECLARATION

I hereby declare that the information provided above is true and accurate to the best of my knowledge and belief.

Pancham Vishwanath Kotian